

Notes: Das, Roberts, and Tybout (Econometrica, 2007)

Market Entry Costs, Producer Heterogeneity, and Export Dynamics

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	Plant-level panel	3
3.2	Observed variables	3
3.3	Export histories and industry patterns	4
4	Models and empirical specifications	4
4.1	Gross potential export profits	4
4.2	Link between profits and export revenues	4
4.3	Dynamic export participation	5
4.4	Estimation strategy	5
4.5	Counterfactual simulations	6
5	Identification and empirical design	6
5.1	What identifies the structural parameters	6
5.2	Initial conditions and selection	6
5.3	What the design can and cannot claim	7
6	Tables and figures	7
6.1	Table I: Posterior parameter distributions	7
6.2	Table II: In-sample model performance	8
6.3	Figure 1: Plant export value and sunk entry costs	8
6.4	Figure 2: Probability of exporting conditional on plant history	9
6.5	Figure 3: Value of exporting and current profit for nonexporting plants	10
6.6	Figure 4: Export response to a correctly perceived 20 percent devaluation	11
6.7	Figure 5: Perceived versus unperceived regime shift	12
6.8	Table III: Export revenue/cost ratios for subsidy plans	13
7	Short summary	13

8	Conclusion	14
8.1	Core conclusion	14
8.2	Economic intuition	14
8.3	Incremental contribution revisited	14
8.4	Policy implications	14
8.5	Limitations	14
8.6	Takeaway	15

1 Big picture

- **Question.** How do plant-level sunk entry costs, producer heterogeneity, and expectations shape export entry, export exit, and the volume response of exports to policy changes?
- **Setting.** The paper studies Colombian manufacturing plants from 1981 to 1991 in three industries: leather products, knitted fabrics, and basic chemicals. These industries are export-oriented enough to observe entry and exit, but differ sharply in technologies, export destinations, and export histories.
- **Core idea.** Exporting is a dynamic decision. A plant that is already exporting can adjust export volume using its existing foreign-market presence, while a nonexporter must first pay a sunk cost to enter foreign markets.
- **Two margins.** Aggregate export responses come from: (i) the intensive margin, where incumbent exporters change export revenues; and (ii) the entry-exit margin, where plants start or stop exporting.
- **History matters.** Previous exporting status changes current incentives because incumbent exporters have already paid the entry cost. This creates hysteresis: plants may continue exporting even when current export profits are low or negative.
- **Expectations matter.** A permanent-looking exchange-rate shift has a larger effect than an unperceived or temporary-looking shift, because entry requires an expected stream of future export profits large enough to cover sunk costs.
- **Heterogeneity matters.** Industries respond differently because plants differ in their expected export profitability and in how many plants are close to their entry threshold.
- **Main simulation result.** A correctly perceived 20 percent devaluation generates a large predicted export response in knitted fabrics, a moderate response in leather products, and a weak response in basic chemicals.
- **Policy takeaway.** Export revenue subsidies are much more cost-effective than subsidies that only reduce entry costs. Revenue subsidies affect all exporters' volume choices, while entry subsidies mainly affect marginal potential entrants.

2 Incremental contribution of the paper

- **From testing sunk costs to measuring them.** Earlier papers used export persistence to infer the presence of sunk export costs. This paper estimates the sunk entry costs in pesos and uses them for counterfactual policy analysis.
- **Joint model of entry and volume.** The paper does not study export participation alone. It models both the discrete decision to export and the continuous decision of how much export revenue to generate.
- **A structural bridge from micro behavior to aggregate export supply.** The model links plant-level profit heterogeneity, entry costs, and expectations to industry-level export responses.
- **Policy evaluation with dynamics.** The paper can ask not only whether exports respond to a devaluation, but how responses differ when producers correctly perceive the regime change versus when they do not.

- **Methodological contribution.** The authors use a Bayesian MCMC estimator for a dynamic type-II-tobit-like model with serially correlated shocks, initial conditions, producer heterogeneity, and selection into exporting.
- **Data-use contribution.** The model is written in terms of variables that are available in standard plant panels: export revenues, domestic revenues, total variable costs, export status, and exchange rates. It avoids requiring plant-level export prices and quantities.
- **Comparison of policy instruments.** The paper directly compares devaluations, revenue subsidies, fixed-cost subsidies, and entry-cost subsidies, showing why policies that affect volumes dominate policies that only subsidize entry.

3 Data and measurement

3.1 Plant-level panel

- The empirical analysis uses Colombian plant-level data originally collected by DANE and cleaned in earlier work by Roberts and Tybout.
- The sample covers plants that operated continuously in the domestic market from 1981 to 1991.
- The three industries are:
 - 32 leather products producers,
 - 40 basic chemicals producers,
 - 64 knitted fabrics producers.
- During the period, Colombia experienced a large real effective exchange-rate depreciation, and all three industries expanded exports.
- Export participation also increased, but the importance of new exporters differed: new entrants generated a large share of later knitted-fabric exports, much less of leather exports, and almost none of basic-chemical exports.

Interpretation: The industry comparison is central. The same macro stimulus can produce different export responses depending on how many plants are close to the export-entry threshold and how much prior exporting experience the industry already has.

3.2 Observed variables

- For each plant and year, the authors observe export participation, export revenues, domestic revenues, and total variable costs.
- They also observe time-invariant plant characteristics, summarized in the baseline model by a size dummy based on initial domestic sales.
- They add a real effective exchange-rate series, modeled as an AR(1) process.
- The data do not contain plant-specific export prices, quantities, export profits, or destination markets.

Interpretation: The model is designed around realistic industrial census data. The absence of prices and quantities is not treated as a fatal weakness; instead, the authors use revenue and cost relationships implied by profit maximization.

3.3 Export histories and industry patterns

- A plant's lagged export status is a key state variable because it determines whether it must pay a sunk entry cost to export this period.
- Incumbent exporters can remain in foreign markets without paying the entry cost again.
- The three industries differ strongly in the stock of incumbent exporters at the start of the period.

Interpretation: This is why the paper emphasizes history. Export growth is not just a response to current exchange rates; it also depends on the inherited distribution of incumbent exporters and potential entrants.

4 Models and empirical specifications

4.1 Gross potential export profits

The paper begins with potential gross export profits for plant i in year t :

$$\ln \pi_{it}^* = \psi_0 z_i + \psi_1 e_t + v_{it},$$

where z_i captures time-invariant plant characteristics, e_t is the log real exchange rate, and v_{it} is a serially correlated plant-specific profit shock.

- A higher exchange rate makes exporting more attractive when $\psi_1 > 0$.
- The disturbance v_{it} captures idiosyncratic demand and cost shocks.
- In estimation, v_{it} is represented as the sum of two independent AR(1) components, allowing a flexible ARMA structure.

Interpretation: This profit equation gives the plant-specific foreign-market payoff before sunk and fixed costs. It is the core object that determines both export volume and export participation.

4.2 Link between profits and export revenues

Because export profits are not observed, the authors use the markup relationship implied by constant-elasticity foreign demand. Potential export profits can be written as a share of potential export revenues:

$$\pi_{it}^* = \eta_i^{-1} R_{it}^{f*}.$$

This implies the export revenue equation:

$$\ln R_{it}^{f*} = \ln \eta_i + \psi_0 z_i + \psi_1 e_t + v_{it}.$$

- η_i is the plant-specific foreign demand elasticity.

- The authors use total variable costs and domestic/export revenue shares to help identify these elasticities.
- Priors on demand elasticities are used because the panel is short and each plant has limited information.

Interpretation: The revenue equation is the bridge from theory to data. It lets the authors infer export profitability from observed revenues and costs rather than from unobserved prices, quantities, or profits.

4.3 Dynamic export participation

Net current export payoffs depend on whether the plant exported in the previous year:

$$u_{it} = \begin{cases} \pi_{it}^* - \gamma_F + \epsilon_{1it}, & y_{it} = 1, y_{i,t-1} = 1, \\ \pi_{it}^* - \gamma_F - \gamma_S z_i + \epsilon_{2it}, & y_{it} = 1, y_{i,t-1} = 0, \\ 0, & y_{it} = 0. \end{cases}$$

- γ_F is a fixed cost of maintaining an export presence.
- $\gamma_S z_i$ is a sunk start-up cost paid by new exporters.
- The sunk cost is paid only when a plant was not exporting in the previous year.

The plant solves the Bellman problem

$$V_{it} = \max_{y_{it} \in \{0,1\}} \{u_{it} + \delta E_t[V_{i,t+1}]\}.$$

Exporting occurs when the current payoff plus the option value of entering or staying in the export market is positive.

Interpretation: The option value is essential. A plant may export today not because today's profits are high, but because exporting preserves the option to continue tomorrow without paying the sunk entry cost again.

4.4 Estimation strategy

- The model is estimated separately for each industry.
- The likelihood combines information from export participation, export revenues, domestic revenues, total variable costs, and the exchange-rate process.
- The likelihood resembles a dynamic type II Tobit model, but it allows serial correlation, endogenous initial export status, plant-level demand elasticities, and nonlinear dynamic selection.
- The authors use a Bayesian Metropolis-Hastings MCMC estimator to characterize the posterior distribution of parameters.

Interpretation: The estimator is computationally demanding, but it lets the paper recover structural parameters that reduced-form export participation regressions cannot recover.

4.5 Counterfactual simulations

After estimating parameters, the authors simulate:

- a correctly perceived permanent 20 percent devaluation,
- an unperceived regime shift in the exchange-rate process,
- changes in exchange-rate volatility,
- revenue subsidies,
- fixed-cost subsidies,
- sunk-entry-cost subsidies.

Interpretation: The counterfactuals show how the same policy affects industries differently because the underlying plant distributions differ.

5 Identification and empirical design

5.1 What identifies the structural parameters

- **Profit-function parameters** are identified by variation in export revenues across exporters and over time, especially as exchange rates and plant shocks change.
- **Demand elasticities** are identified using the relationship between total variable costs, domestic revenues, export revenues, and the share of revenues generated abroad.
- **Sunk entry costs** are identified from state dependence in export participation: among plants with similar expected profit streams, previous exporters behave differently from previous nonexporters.
- **Fixed costs** are identified from exit behavior among plants with positive expected export profits.
- **Shock processes** are identified from the serial behavior of export revenues and participation.

Interpretation: The identifying logic separates true state dependence from persistent unobserved profitability. Sunk costs generate a discontinuity between incumbent exporters and potential entrants that cannot be explained by serial correlation alone.

5.2 Initial conditions and selection

- Initial export status is endogenous because it reflects pre-sample profitability and entry choices.
- The authors use a Heckman-style reduced-form initial-conditions equation rather than simulating the full steady-state distribution.
- The export revenue equation is only observed for exporters, so the estimator jointly models participation and revenue to address selection.

Interpretation: The paper's most important econometric problem is that observed exporters are selected. The one-step structural likelihood is intended to use both participation and revenue information consistently.

5.3 What the design can and cannot claim

- The paper is not a randomized experiment. Its counterfactuals are model-based exercises.
- The credibility of the policy simulations depends on the maintained assumptions about demand, costs, shock processes, and export-entry costs.
- The sample contains only three industries and only plants that remain continuously active in the domestic market.
- The model treats the plant as the decision-making unit because the data do not link plants within multi-plant firms.

Interpretation: The contribution is not causal identification from an exogenous shock alone. It is a disciplined structural quantification of how export dynamics work under specified assumptions.

6 Tables and figures

6.1 Table I: Posterior parameter distributions

Read:

- Table I reports posterior means and standard deviations for the structural parameters in each industry.
- Exchange-rate coefficients are large in leather products and knitted fabrics, but much smaller in basic chemicals.
- The posterior roots of the export-profit shock process show substantial persistence.
- Sunk entry costs are large: roughly 50 to 64 million 1986 pesos depending on industry and plant size class.
- Fixed costs are close to zero on average, although the cost shocks imply that fixed costs can matter for some plants in some years.
- Foreign demand elasticities are high, implying relatively small gross profit shares of export revenue.

Economic message: Table I is the paper’s structural foundation. It shows that the export market is costly to enter, profit shocks are persistent, and exchange-rate sensitivity varies sharply across industries.

TABLE I
POSTERIOR PARAMETER DISTRIBUTIONS (MEANS AND STANDARD DEVIATIONS)

	Leather Products	Basic Chemicals	Knitted Fabrics	Priors
Profit Function Parameters				
ψ_{01} (intercept)	-13.645 (4.505)	1.143 (3.642)	-12.965 (3.058)	$\psi_{01} \sim N(0, 500)$
ψ_{02} (domestic size dummy)	1.544 (0.789)	1.862 (0.813)	1.362 (0.449)	$\psi_{02} \sim N(0, 500)$
ψ_1 (exchange rate coefficient)	4.323 (0.957)	0.975 (0.745)	4.047 (0.640)	$\psi_1 \sim N(0, 500)$
λ_1^1 (root, first AR process)	0.787 (0.180)	-0.383 (0.186)	0.458 (0.258)	$\lambda_1^1 \sim U(-1, 1)$
λ_1^2 (root, second AR process)	0.952 (0.018)	0.951 (0.022)	0.709 (0.103)	$\lambda_1^2 \sim U(-1, 1)$
$\sigma_{\varepsilon_1}^2$ (variance, first AR process)	0.282 (0.144)	0.320 (0.109)	0.469 (0.250)	$\ln(\sigma_{\varepsilon_1}^2) \sim N(0, 20)$
$\sigma_{\varepsilon_2}^2$ (variance, second AR process)	0.422 (0.146)	0.491 (0.137)	0.809 (0.264)	$\ln(\sigma_{\varepsilon_2}^2) \sim N(0, 20)$
ν (foreign elasticity premium)	-0.016 (0.022)	0.849 (0.126)	0.950 (0.047)	$\nu \sim U(-1, 1)$
λ_ξ (root, measurement error)	0.336 (0.070)	0.962 (0.011)	0.935 (0.013)	$\lambda_\xi \sim U(-1, 1)$
σ_ξ (std. error, ξ innovations)	0.011 (0.001)	1.277 (0.389)	1.312 (0.264)	$\ln(\sigma_\xi) \sim N(0, 20)$
Foreign Demand Elasticities (quintiles only)				
η_{Q1} (demand elasticity, quintile 1)	8.020 (2.907)	12.098 (13.881)	10.289 (12.032)	$\ln(\eta - 1) \sim N(2, 1)$
η_{Q2} (demand elasticity, quintile 2)	12.282 (13.351)	12.974 (18.682)	12.314 (8.330)	$\ln(\eta - 1) \sim N(2, 1)$
η_{Q3} (demand elasticity, quintile 3)	17.866 (11.089)	14.139 (13.363)	13.780 (16.725)	$\ln(\eta - 1) \sim N(2, 1)$
η_{Q4} (demand elasticity, quintile 4)	37.189 (25.331)	24.604 (27.253)	36.279 (32.844)	$\ln(\eta - 1) \sim N(2, 1)$
Dynamic Discrete Choice Parameters				
γ_1 (sunk cost, size class 1)	63.690 (1.934)	62.223 (3.345)	61.064 (2.628)	$\gamma_1 \sim N(0, 500)$
γ_2 (sunk cost, size class 2)	52.615 (4.398)	50.561 (5.043)	59.484 (2.361)	$\gamma_2 \sim N(0, 500)$
γ_F (fixed cost)	-0.610 (1.042)	1.635 (0.983)	1.372 (1.340)	$\gamma_F \sim N(0, 500)$
σ_{ε_1} (std. error, ε_1)	12.854 (6.171)	7.517 (4.109)	32.240 (8.382)	$\ln(\sigma_{\varepsilon_1}) \sim N(0, 20)$
σ_{ε_2} (std. error, ε_2)	30.627 (7.831)	32.432 (3.196)	17.630 (4.737)	$\ln(\sigma_{\varepsilon_2}) \sim N(0, 20)$
Initial Conditions Parameters				
α_0 (intercept)	-3.559 (6.523)	-13.693 (7.069)	-40.811 (21.379)	$\alpha_0 \sim N(0, 500)$
α_1 (domestic size dummy)	16.484 (9.965)	25.868 (11.959)	23.397 (14.762)	$\alpha_1 \sim N(0, 500)$
$\alpha_2(x_1)$	29.388 (11.675)	-18.028 (11.638)	31.603 (18.165)	$\alpha_2 \sim N(0, 500)$
$\alpha_3(x_2)$	3.451 (4.861)	8.908 (5.710)	16.561 (15.519)	$\alpha_3 \sim N(0, 500)$

Table I: posterior parameter distributions.

6.2 Table II: In-sample model performance

Read:

- Table II compares actual entry, exit, and export participation rates with simulated model predictions.
- Predicted average entry rates are close to actual rates: 0.038 versus 0.041 for knitted fabrics, 0.047 versus 0.041 for leather products, and 0.034 versus 0.028 for basic chemicals.
- Predicted export participation is also close to actual participation: 0.170 versus 0.163 in knitted fabrics, 0.592 versus 0.585 in leather products, and 0.309 versus 0.325 in basic chemicals.
- The model also performs reasonably across export revenue quartiles.

Economic message: The model is not only a theoretical structure; it approximately reproduces the observed entry, exit, and participation patterns that matter for the counterfactuals.

6.3 Figure 1: Plant export value and sunk entry costs

Read:

- Figure 1 compares each plant's expected value of exporting with its sunk entry cost.
- Plants are sorted by type and expected export value, separately by industry.
- Many small plants have expected export values below entry costs.
- Knitted fabrics show relatively clustered plants, while leather and basic chemicals show more heterogeneity among larger plants.
- The location of plants relative to the entry threshold explains why entry can be important in knitted fabrics and less important in basic chemicals.

Economic message: The figure gives the main intuition for heterogeneous responses: export booms occur on the entry margin only when many plants sit near their entry threshold.

TABLE II
ENTRY RATES, EXIT RATES, AND EXPORT MARKET PARTICIPATION RATES (10TH AND 90TH PERCENTILE BOUNDS IN PARENTHESES)

	Knitting Mills		Leather Products		Basic Chemicals	
	Predicted (lower, upper)	Actual	Predicted (lower, upper)	Actual	Predicted (lower, upper)	Actual
Exporter Turnover Rates						
Entry	0.038 (0.025, 0.050)	0.041	0.047 (0.031, 0.063)	0.041	0.034 (0.020, 0.050)	0.028
Exit	0.025 (0.016, 0.036)	0.020	0.028 (0.016, 0.044)	0.025	0.040 (0.028, 0.054)	0.035
Export Market Participation Rates						
Exporting	0.170 (0.118, 0.235)	0.163	0.592 (0.131, 0.278)	0.585	0.309 (0.254, 0.365)	0.325
Broken down by export revenue quartiles						
First quartile	0.047 (0.023, 0.071)	0.036	0.203 (0.131, 0.278)	0.139	0.097 (0.059, 0.140)	0.073
Second quartile	0.047 (0.023, 0.071)	0.040	0.131 (0.093, 0.175)	0.145	0.081 (0.055, 0.109)	0.086
Third quartile	0.031 (0.020, 0.053)	0.038	0.131 (0.093, 0.175)	0.142	0.061 (0.040, 0.078)	0.080
Fourth quartile	0.042 (0.028, 0.063)	0.046	0.114 (0.069, 0.162)	0.156	0.069 (0.040, 0.098)	0.091

(a) Table II: in-sample fit for entry, exit, and participation.

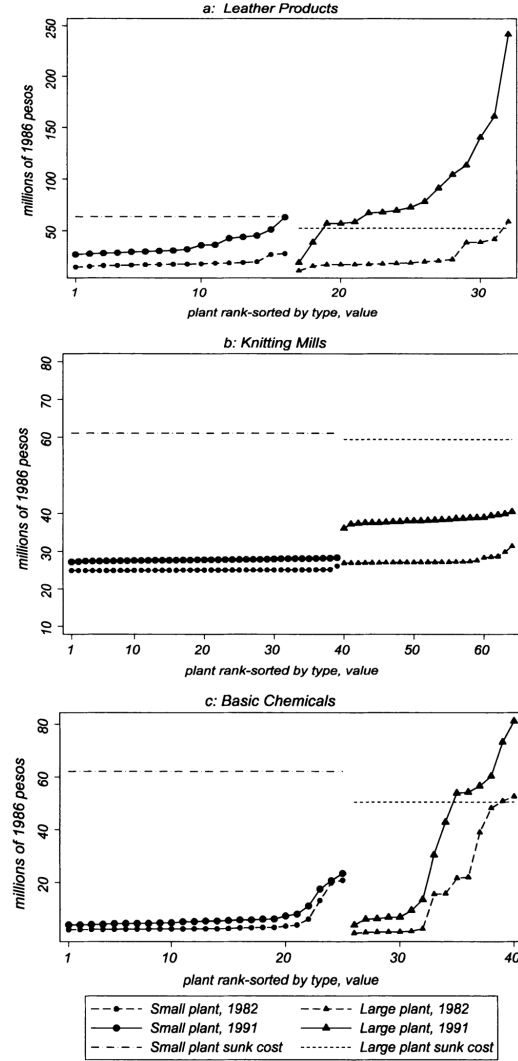


FIGURE 1.—Plant export value and sunk entry costs.

(b) Figure 1: expected export value versus sunk entry costs.

6.4 Figure 2: Probability of exporting conditional on plant history

Read:

- Figure 2 plots two probabilities for each plant: the probability of remaining in the export market if already exporting, and the probability of entering if not already exporting.
- The probability of remaining an exporter is generally much higher than the probability of entering.
- This gap is especially large for plants with low expected export value.
- The figure visually demonstrates export hysteresis: incumbency in the export market matters.

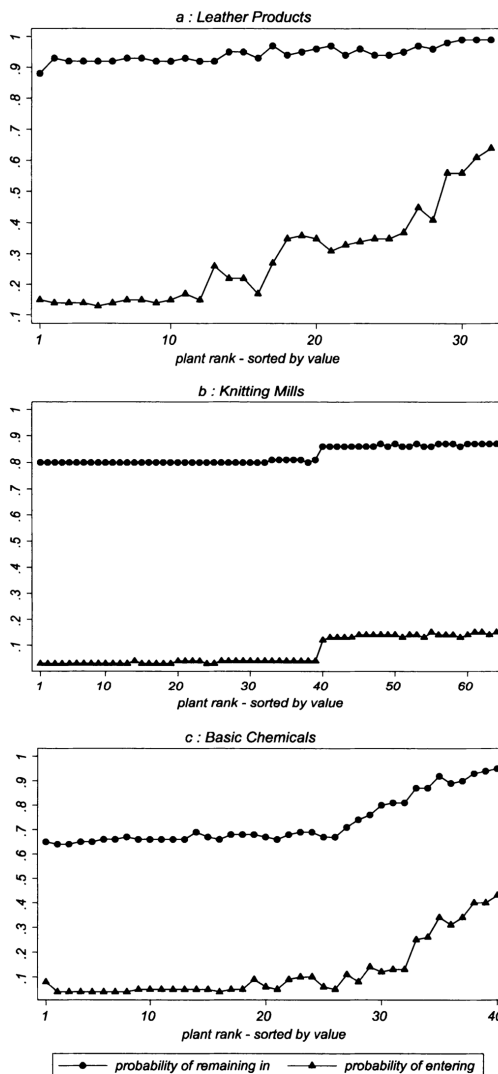
Economic message: History matters because entry is costly. The same plant can be likely to continue exporting if already in the market but unlikely to enter if currently out.

6.5 Figure 3: Value of exporting and current profit for nonexporting plants

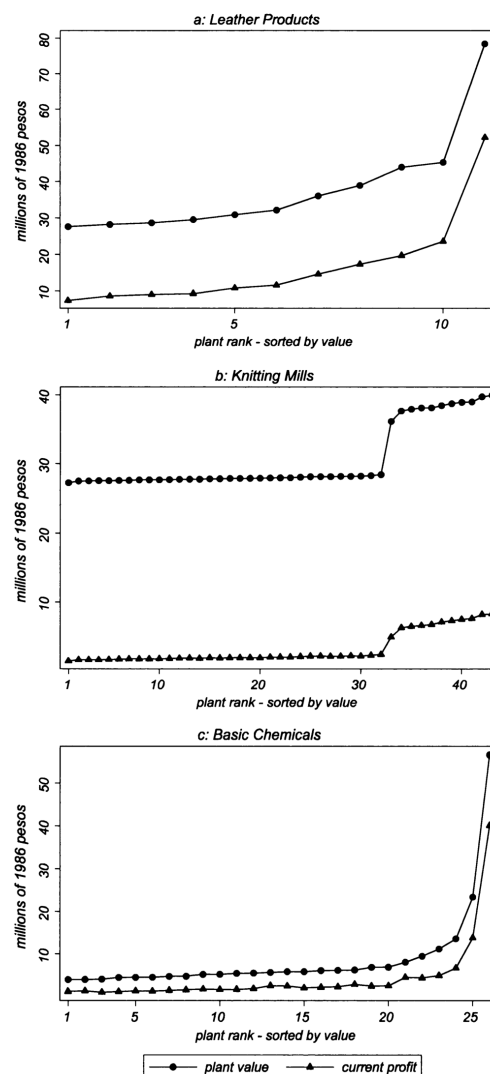
Read:

- Figure 3 compares the value of exporting with current export profit for nonexporting plants.
- For many plants, the value of exporting is much larger than current profit.
- This gap is the option value of becoming an exporter and then avoiding future reentry costs.
- Option values are especially important in knitted fabrics.

Economic message: A static model based only on current profits would miss a major part of the export-entry incentive. Dynamic option values are central.



(a) Figure 2: export probabilities conditional on plant history.



(b) Figure 3: value of exporting and current profit for nonexporters.

6.6 Figure 4: Export response to a correctly perceived 20 percent devaluation

Read:

- Figure 4 simulates a permanent 20 percent depreciation in the steady-state exchange rate, correctly perceived by plants.
- The number of exporters rises by about 37 percent in knitted fabrics, 14 percent in leather products, and 4 percent in basic chemicals over ten years.
- Export revenues rise by about 108 percent in knitted fabrics, 68 percent in leather products, and 14 percent in basic chemicals.
- Revenue responses exceed exporter-count responses because incumbent exporters also expand export volumes.

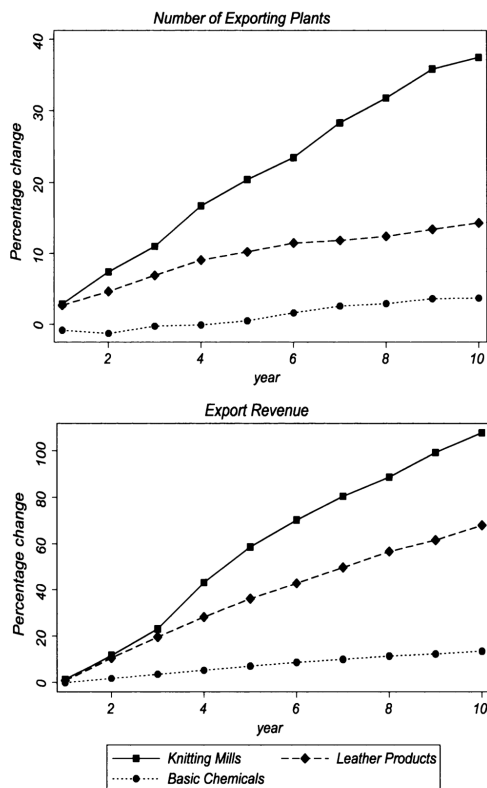
Economic message: The same devaluation generates very different industry responses. Knitted fabrics respond strongly because both entry and volume margins are active; basic chemicals respond weakly because few plants are near the entry margin.

6.7 Figure 5: Perceived versus unperceived regime shift

Read:

- Figure 5 repeats the devaluation experiment for knitted fabrics, comparing a perceived regime shift with an unperceived regime shift.
- When firms understand that the exchange-rate process has changed, entry and export revenues rise more.
- When firms continue to believe the old process governs future exchange rates, the response is much smaller on the entry margin.
- The difference reflects the role of expectations in covering sunk entry costs.

Economic message: Credibility matters. A policy that changes current profits but is not believed to be persistent has much less power to induce new export entry.



(a) Figure 4: correctly perceived 20 percent devaluation.

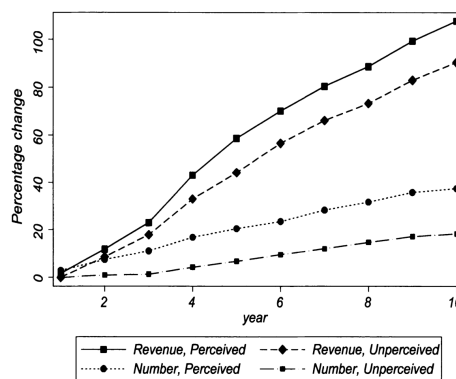


FIGURE 5.—Response to a perceived versus unperceived regime shift: 20 percent steady state devaluation, knitting mills.

(b) Figure 5: perceived versus unperceived regime shift.

6.8 Table III: Export revenue/cost ratios for subsidy plans

Read:

- Table III reports export-revenue gains per peso of subsidy cost, five years after policy introduction.
- Revenue subsidies produce the highest benefit-cost ratios: from 7.52 to 19.23 depending on industry and subsidy rate.
- Fixed-cost subsidies are weaker, generating no more than 3.96 pesos of extra export revenue per peso spent.
- Entry-cost subsidies perform worst and can even generate negative ratios in leather products and basic chemicals.
- Entry-cost subsidies reduce the cost of reentry and can weaken the option value of staying in the export market, increasing turnover.

Economic message: Subsidizing export volume is more effective than subsidizing entry. The biggest export gains come from policies that affect incumbent exporters, not only marginal potential entrants.

TABLE III
EXPORT REVENUE/COST RATIOS FOR ALTERNATIVE SUBSIDY PLANS
(MEANS OVER 300 SIMULATIONS)

	Knitting Mills	Leather Products	Basic Chemicals
Revenue Subsidies			
2 percent	19.23	11.81	11.17
5 percent	15.67	9.94	9.56
10 percent	10.09	7.74	7.52
Entry Cost Subsidies			
25 percent	2.03	-0.54	-0.068
50 percent	1.02	-0.67	-0.49
100 percent	0.14	-0.26	-0.23
Fixed Cost Subsidy			
2 million pesos	3.96	2.22	2.66
10 million pesos	0.99	0.68	0.69

Table III: export revenue/cost ratios for alternative subsidy plans.

7 Short summary

- The paper develops and estimates a dynamic structural model of export supply.
- Plants choose whether to export and how much export revenue to generate.
- Exporting involves sunk entry costs, fixed costs, persistent profit shocks, plant heterogeneity, and expectations about future exchange rates.
- The empirical setting is Colombian manufacturing from 1981 to 1991 in leather products, knitted fabrics, and basic chemicals.
- Sunk entry costs are large, around 50 to 64 million 1986 pesos across industries and size classes.
- Incumbent exporters are much more likely to remain exporters than nonexporters are to enter.

- Knitted fabrics are more responsive on the entry margin because many plants are near their entry thresholds.
- Leather products and basic chemicals are more heterogeneous, so volume adjustment by existing exporters is more important.
- A correctly perceived 20 percent devaluation produces large export growth in knitted fabrics and smaller effects elsewhere.
- Revenue subsidies dominate entry-cost subsidies in benefit-cost terms.

8 Conclusion

8.1 Core conclusion

Das, Roberts, and Tybout show that export supply is dynamic and micro-heterogeneous. Aggregate export responses cannot be understood only from current exchange rates or average industry conditions. They depend on sunk entry costs, prior export status, expectations, and the distribution of plant-level export profitability.

8.2 Economic intuition

The central intuition is that exporting creates a state variable. Once a plant pays the cost of entering a foreign market, it owns an option to keep exporting in later periods without paying that cost again. This option makes current export status persistent and makes expectations about future profitability important. When many plants are close to the entry threshold, a modest policy change can trigger large entry. When few plants are close to that threshold, the same policy mostly affects the export volume of incumbent exporters.

8.3 Incremental contribution revisited

The paper advances the export dynamics literature in three ways. First, it quantifies market entry costs rather than merely documenting export persistence. Second, it jointly models export participation and export revenues. Third, it uses the structural model to compare policy instruments, showing that the ranking of policies depends on the dynamic microstructure of exporting.

8.4 Policy implications

Export promotion policies should be evaluated by the margin they affect. Policies that reduce entry costs mainly help marginal nonexporters and can have weak or even adverse effects if they reduce the option value of staying in export markets. Policies that subsidize export revenues affect incumbent exporters as well as entrants and therefore generate larger export revenue per peso spent.

8.5 Limitations

The estimates are based on three Colombian industries and on plants continuously active in the domestic market. The model abstracts from simultaneous domestic entry/exit, plant links within

multi-plant firms, destination-specific markets, and direct price/quantity measurement. The counterfactuals are therefore best read as structural simulations under explicit assumptions rather than as reduced-form causal estimates.

8.6 Takeaway

The main lesson is that export booms are shaped by the micro distribution of firms. Sunk costs and expectations can make export supply sluggish, but when many firms are near the threshold, the same forces can also generate sharp waves of entry. For policy, the paper's strongest message is that subsidizing the volume margin is usually more powerful than subsidizing entry alone.